



MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

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Chair of Empirical Microeconomics

ADVANCED MICROECONOMICS

General Equilibrium II - Welfare Economics

Winter Term 2025/2026

Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Duality (L4+L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Safety Regulations and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

Literature

- Autor, David (2016): Lecture Notes 11
- Varian (2022): Chapter 33
- Jensen, Robert (2007): “The digital divide: Information (technology), market performance and welfare in the South Indian fisheries sector” *The Quarterly Journal of Economics*, 122(3): 879–924

Overview

- 1 Welfare Economics and General Equilibrium
- 2 What is Welfare?
- 3 The First Welfare Theorem
- 4 The Second Welfare Theorem
- 5 Policy Implications
- 6 Application: The role of information (Jensen 2007)
- 7 Summary

Recap: General Equilibrium Theory

- **General equilibrium analyzes** *all* markets simultaneously and their interactions.
- In an **exchange economy**, endowments and preferences fully determine feasible allocations.
- The **Edgeworth Box** contains all feasible allocations, and indifference curves and thus visualizes all Pareto improvements and **Pareto-efficient allocations** (contract curve).
- **Pareto efficiency**: no consumer can be made better off without making the other worse off.

$$MRS_A = MRS_B$$

Recap: General Equilibrium Theory

- A **Walrasian (competitive) equilibrium** is a set of prices and allocations where:
 - both consumers maximize utility given prices,
 - all markets clear (zero excess demand).
- Under convex preferences and perfect markets, a Walras equilibrium exists and is **Pareto efficient**.
- GET answers:
 - How do markets reach equilibrium?
 - What do equilibria look like (prices, allocations)?
 - Are these equilibria efficient?

From General Equilibrium to Welfare Economics

- But GET leaves open:
 - Is every competitive equilibrium socially desirable?
 - What is the role of equity vs. efficiency?
- **Welfare Economics** provides the bridge:
 - uses social and Pareto efficiency concepts,
 - links competitive equilibria with efficient allocations,
 - clarifies the relationship between markets and social optimality.

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What is “Welfare” in Welfare Economics?

Welfare

Welfare measures of aggregate well-being in the economy

- **Utilitarianism** (Bentham, Mill) - Welfare as *sum of utilities*:

$$U = \sum_{i=1}^N u_i$$

- **Measurement problems:** interpersonal comparisons of cardinal utility are necessary

Welfare Concepts: A Brief History

- **Pigou/Marshall**: economic welfare = national dividend, consumers' surplus.
- **Pareto** (1906): welfare without utility comparisons → Pareto criterion.
- **Hicks-Kaldor** (1939/41): compensation principle within the Pareto framework (Winner compensates loser and is still better off)
→ Remains the operative policy criterion in economics
- **Bergson-Samuelson** (1938/47): social welfare functions (explicit ethical value judgments).
- **Arrow** (1951): impossibility of aggregating preferences consistently (i.e. building social welfare functions)
- **Modern approaches**: happiness (Easterlin), non-welfarism - rights, capabilities, environment, freedom to choose lifestyle (Sen) ...

Why GE Theory Uses the Pareto Concept

- Pareto criterion is a minimal **value-judgment free** concept:

Welfare = Pareto Welfare

An allocation is welfare-improving if at least one individual is better off and no one is worse off.

- Only provides guidance about ‘improvements’ from a given position.
- No interpersonal utility comparisons.
- No statement about fairness, distribution, or happiness.
- Makes welfare analysis possible within positive economics.
- Allows strong statements about competitive equilibria (**Welfare Theorems**)
- But also very limiting:
 - says nothing about fairness, equity, poverty.
 - ignores broader ethical concerns.

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The first fundamental theorem of Welfare Economics

First Welfare Theorem

If (P_1, P_2) and (X_A, X_B) are a **Walrasian equilibrium** and consumer preferences are strictly monotone, this allocation will be **Pareto efficient**.

- Theorem goes back to **Leon Walras** and implies that the market will lead to a Pareto efficient allocation (if perfect market conditions hold).
- Free market ensures that all potential gains from trade will be exhausted, while nobody will be made worse off compared to the initial situation.
- Maximize utility for all individuals while
 - 1 not making anyone worse off.
 - 2 reaching a point in which no more improvements are possible (make “the pie” as large as possible).
 - 3 do not waste available resources (market clearance).

The first fundamental theorem of Welfare Economics

Relevance

- The free market will **maximize the pie** (lead to an efficient allocation) under ideal circumstances.
- A central planner could do this only with a lot of information (e.g. on individual preferences)!
- **BUT**: there are strong assumptions that have to be fulfilled
 - no transaction costs
 - no market power
 - no externalities
 - full information
- **Plus** the First Theorem does not say anything about **equity** associated with the resulting allocation.

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The second fundamental theorem of Welfare Economics

Second Welfare Theorem

If market conditions hold, **every Pareto efficient allocation** of resources can be reached via the free market if **redistribution of initial allocation** is possible without costs.

- The theorem implies that there does not need to be an efficiency-equity trade off and that it is possible to **separate the efficiency and equity aspects of economic outcomes**, suggesting that governments can intervene to redistribute resources without compromising overall efficiency.
- Even when markets fail to allocate resources equitable, there is still potential for achieving equity through government-led redistribution.
- Interventions aimed at correcting inequalities can coexist with market-based solutions.

The second fundamental theorem of Welfare Economics

*Imagine two farmers. One has plenty of seeds but very little land, while the other has abundant land but only a small amount of seeds. If we simply introduce **market prices** without changing the initial endowments, the resulting competitive equilibrium will be pareto-efficient but will not necessarily reflect a “fair” allocation. However, suppose a **planner first redistributes** some of the land to the first farmer and some of the seeds to the second farmer. This creates a more balanced starting point. Once this redistribution of endowments has been made, the competitive market mechanism can take over. Given appropriate prices, trade between the two farmers will lead them to an allocation on the Pareto-efficient contract curve.*

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Policy Implications

The two fundamental theorems imply

- that **policy interventions** cannot lead to Pareto improvements and are thus not warranted from efficiency point of view if markets fulfill “**perfect market conditions**” (first theorem).
- But interventions could be considered based on **equity-considerations**.
- The second theorem implies that the market can lead to any efficient allocation, after **re-allocation of initial resources**, i.e., policy-makers should aim at “re-allocating initial resources” (through lump-sum transfers).
- Edgeworth Box shows that leaving allocation of initial resources untouched and instead **changing prices** (i.e., intervening in the market mechanism) may not result in an efficient allocation.

Should policy never distort prices?

- No, there are many markets in which (at least one of the) four **market conditions do not hold!**
 - E.g.: Externalities, not full information,...
- If the market conditions do not hold, the market allocation is not necessarily efficient (the first theorem doesn't apply) and thus government intervention to change prices (*Pigou taxes, e.g. CO2-tax*) can improve efficiency.
- The different situations in which government intervention is warranted are studied in depth in the field of **public economics**.

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Jensen 2007

THE QUARTERLY JOURNAL OF ECONOMICS

Vol. CXXII

August 2007

Issue 3

THE DIGITAL PROVIDE: INFORMATION (TECHNOLOGY),
MARKET PERFORMANCE, AND WELFARE IN THE
SOUTH INDIAN FISHERIES SECTOR*

ROBERT JENSEN

Jensen 2007

- We will discuss the following questions:
 - What is the paper about?
 - How is it related to our class?
 - What data and methods are used?
 - What are the paper's (main) results?
 - How valid are the paper's findings?
 - What is your personal take-away from this paper?

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Summary

- Welfare economics builds on GET and uses the Pareto concept to evaluate allocations without interpersonal utility comparisons.
- The **First Welfare Theorem** states that, under perfect market conditions and convex preferences, every competitive equilibrium is Pareto efficient.
- The **Second Welfare Theorem** states that any Pareto-efficient allocation can be decentralized as a competitive equilibrium after suitable redistribution of initial endowments.
- This implies a separation between **efficiency** (achieved through markets) and **equity** (achieved through redistribution).
- When perfect market conditions fail (e.g., externalities, asymmetric information), market outcomes need not be efficient, and corrective interventions may improve welfare.
- Jensen (2007) shows empirically that improved information (mobile phones) leads to better market functioning and measurable gains in welfare.

Important Keywords

- 1 Welfare Economics
- 2 Pareto Efficiency
- 3 Pareto Improvement
- 4 Social Welfare
- 5 First Welfare Theorem
- 6 Second Welfare Theorem
- 7 Efficiency–Equity Trade-off
- 8 Endowment Redistribution
- 9 Competitive Equilibrium
- 10 Perfect Market Conditions
- 11 Market Failure
- 12 Information and Market Performance

Questions?

